Customer Churn Prediction Using Machine Learning

Building intelligent models to predict, prevent, and reduce customer churn.

TEAM MEMBERS

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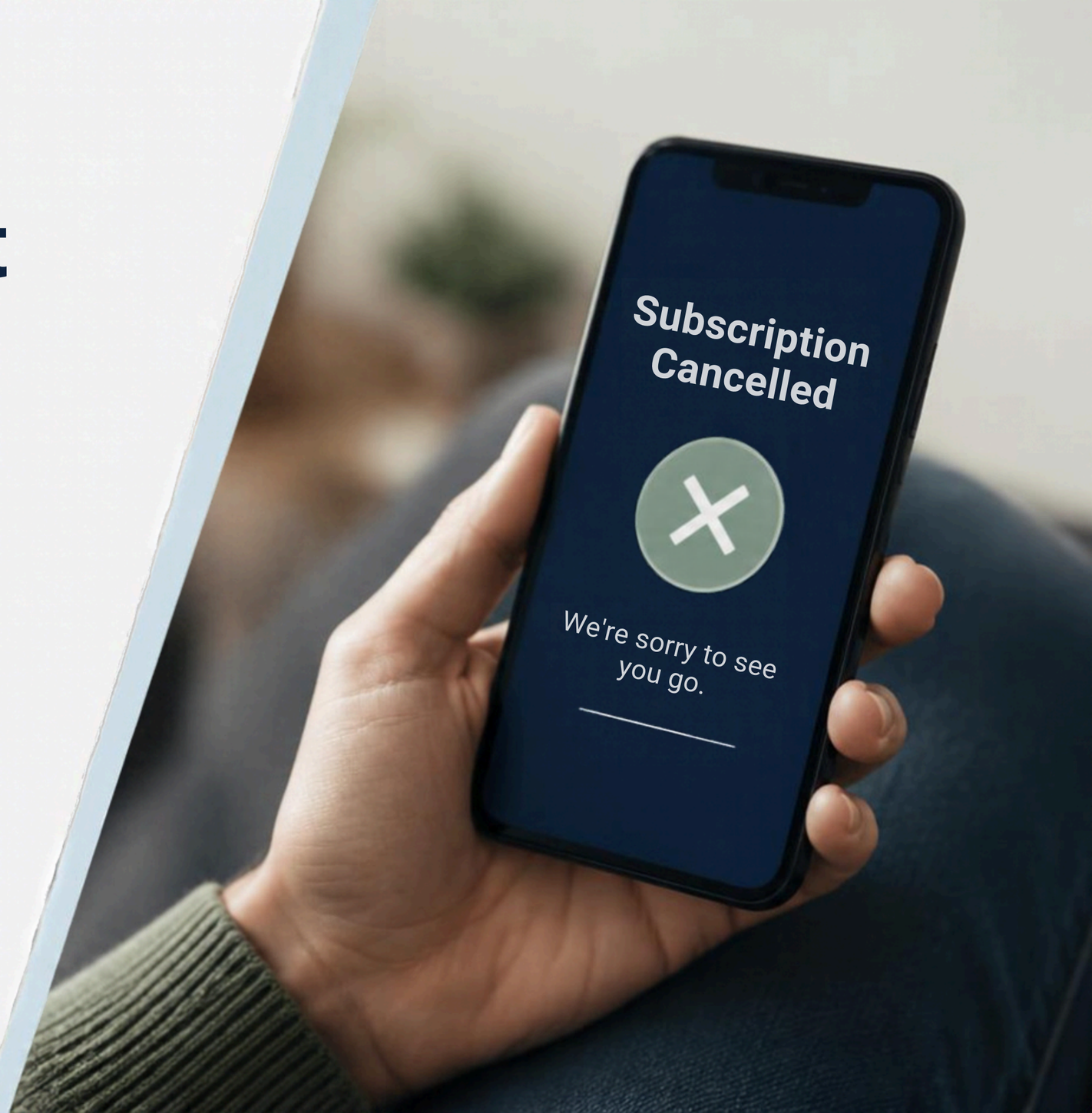
Introduction & Problem Statement

Customer churn happens when users stop using a service or cancel their subscription.

It leads to revenue loss and higher customer acquisition costs.

Early prediction helps identify at-risk customers before they leave.

Telecom subscriptions are especially vulnerable to churn.



Objectives & Proposed Solution

An end-to-end machine learning approach to predict and prevent customer churn.

Understand the Data

Analyze historical customer and subscription data to uncover churn patterns.

Build and Compare Models

Train multiple classification algorithms and compare their performance.

Validate Performance

Evaluate the best model on unseen data to ensure reliability and generalization.

Predict Churn Risk

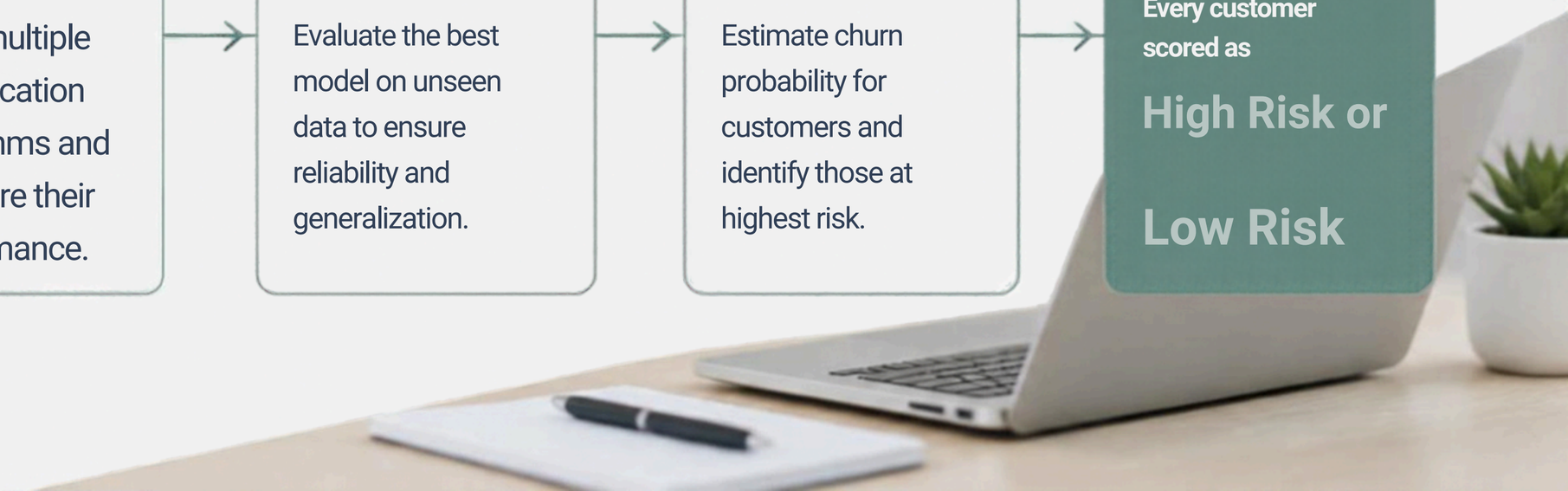
Estimate churn probability for customers and identify those at highest risk.

Actionable Output

Every customer scored as

High Risk or

Low Risk



Dataset Description

IBM Telco Customer Churn data

Approximately
7,043 records

Around
21 columns

Key features tenure, Contract, MonthlyCharges, TotalCharges, InternetService, TechSupport

Target variable Churn Yes/No

Methodology & Data Preprocessing

A clean, structured pipeline ensures high-quality data for reliable churn prediction.

01

Load & Inspect Data

Import the dataset using Pandas and explore structure, types, and summary statistics.

02

Clean & Prepare Data

Handle missing values, fix data types, and remove inconsistencies.

03

Encode Features

Convert categorical variables into machine-readable formats.

04

Split Data for Modeling

Separate features and target, then create train and test sets for evaluation.

Machine Learning Models

We evaluate interpretable models for supervised binary classification.

Logistic Regression

Linear model that estimates the probability of churn.

Provides a simple, interpretable baseline.

Decision Tree

Splits data using feature rules to predict churn.

Easy to interpret and explain.

Random Forest

Ensemble of decision trees that improves accuracy and reduces overfitting.

More robust generalization.

Model Comparison Metrics

Precision	Among predicted churners, the proportion who actually churned.
Recall	Among actual churners, the proportion correctly identified.
F1-Score	Harmonic mean of precision and recall for balanced evaluation.
Accuracy	Overall proportion of correct predictions.

System Architecture

End-to-end ML pipeline that learns from historical data and predicts churn for new customers.

Historical Data

Collect past customer data and churn labels.

Data Preprocessing

Clean, transform, and prepare data for modeling.

Model Training

Train machine learning models on prepared data.

Model Evaluation

Evaluate performance and validate the best model.

New Customer Input

Input new customer data into the trained model.

Churn Prediction

Predict churn probability and classify customer risk.

Tools & Technologies

Python

Core programming language

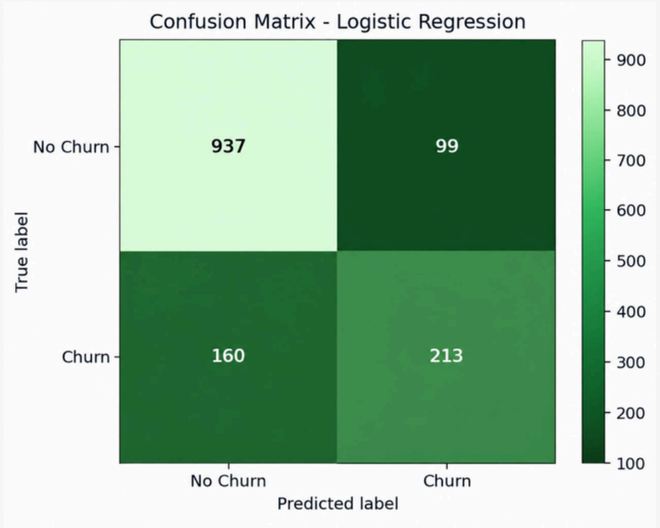
Google Colab

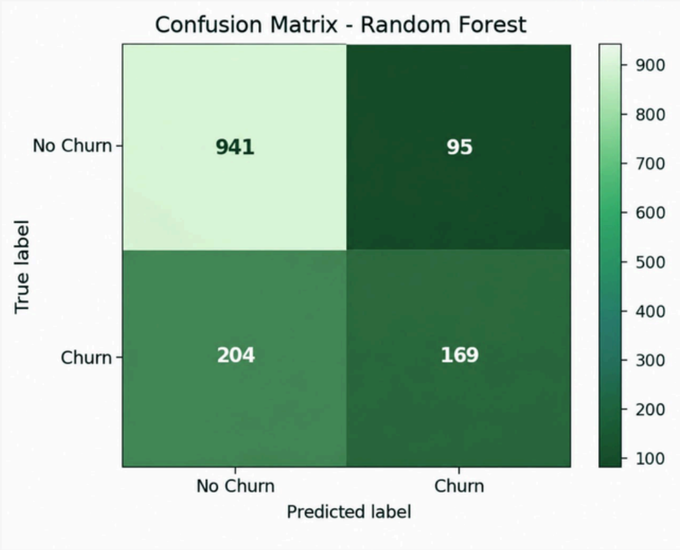
Cloud environment for development

Pandas, NumPy, Scikit-Learn

Data processing and machine learning

Model Evaluation & Results

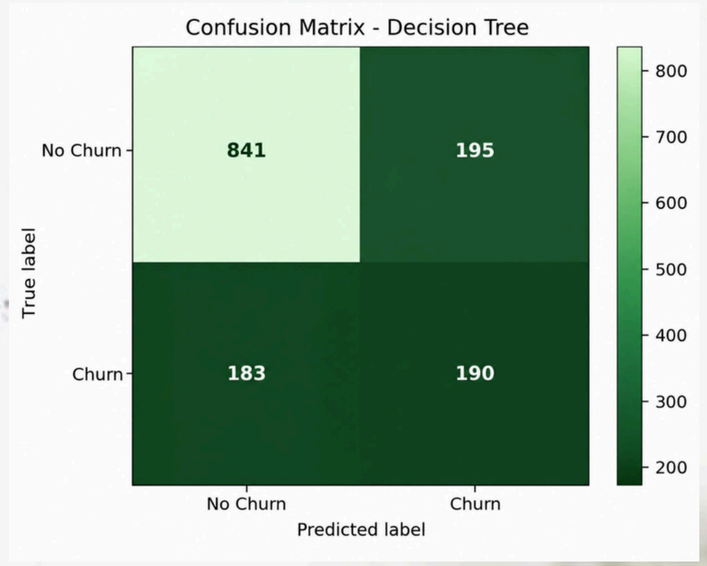




MODEL PERFORMANCE SUMMARY

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Logistic Regression	81.62	68.27	57.10	62.19
Random Forest	78.78	64.02	45.31	53.06
Decision Tree	74.17	51.18	52.28	51.72
Best Model	**81.62**	**68.27**	**57.10**	**62.19**

CONFUSION MATRIX



Inference, Applications & Future Scope

INFERENCE	Estimates churn risk for new customers based on historical behavior patterns.
APPLICATIONS	Useful for telecom, subscription services, CRM systems, and customer retention strategies.
IMPROVEMENTS	Enhance with richer data, feature engineering, and model refinement.
FUTURE SCOPE	Develop web interface, deploy in production, and enable continuous monitoring for performance.

Conclusion & References

Machine learning enables us to identify churn risk early and take action to retain valuable customers.

IBM Telco Customer Churn sample dataset

Scikit-learn documentation

Pandas documentation

NumPy documentation

Thank You.